

Elia Jean Botros

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EDUCATION

The Pennsylvania State University, University Park

Expected Graduation: December 2026

Bachelor of Science in Industrial Engineering

GPA: 3.3

Minor: Engineering Leadership Development

SKILLS & CERTIFICATIONS

Software and General Skills: MATLAB, Python, SQL, Excel, SolidWorks, AutoCAD, Simio, Prototwin, Microsoft Project, Minitab, LINDO, Figma, Process Flow Mapping, Time & Motion Studies

Certification: Lean Six Sigma Green Belt - IISE (Expected End of Sept. 2026)

WORK EXPERIENCE

Kew Gardens Animal Hospital, Queens, New York

May 2026 - Aug. 2026

Animal Hospital Assistant

- Applied process standardization to surgical prep workflows, reducing average instrument setup time and improving procedure throughput across a high-volume clinical environment.
- Designed and maintained standardized equipment and inventory protocols.
- Optimized appointment scheduling and patient intake processes to balance daily caseload, reduce bottlenecks, and minimize patient wait times.

ENGINEERING PROJECTS

Cardinal Systems Automation Feasibility Study, Schuylkill Haven, PA

Aug. 2026 - Present

Project Lead & Process Analyst

- Coordinating directly with the sponsor's President, CFO, and Director of Operations to define project scope and deliverable milestones.
- Conducted on-site time studies across manual work stations, breaking the process into work elements to establish a cycle time baseline and identify bottleneck operations.
- Evaluating integration feasibility of four idle Yaskawa Motoman GP180 robots against reach, payload, and layout constraints while modeling cost and payback versus the current manual line.

ACRP University Design Competition

Jan. 2025 - May 2025

Team Lead & System Designer

- Led system design of GSE Sync, an AI-powered fleet routing system using reinforcement learning and shortest-path algorithms to optimize 250+ ground support vehicles at Seattle-Tacoma International Airport, targeting a 60% reduction in vehicle idle time.
- Collaborated with a 4 member team to develop MATLAB and SUMO simulation prototypes using real OpenStreetMap airport data to model dynamic vehicle routing across multiple flight delay scenarios.
- Performed risk analysis and developed mitigation strategies across identified risk categories.

LEADERSHIP & INVOLVEMENT

Mentor & Tutor, FIRST Robotics Competition Team 2638

Jan. 2024 - May 2024

- Provided remote technical mentorship to FIRST Robotics Team 2638, advising on mechanical systems design, CAD modeling in SolidWorks, and systems-level integration.
- Guided students through engineering design tradeoffs, manufacturing feasibility, and competition requirements through virtual collaboration.

Broadcast Lead, Penn State Intramural Sports

Jan. 2023 - Oct. 2023

- Led live broadcast operations coordinating technical setup, production flow, and real-time problem solving during events.